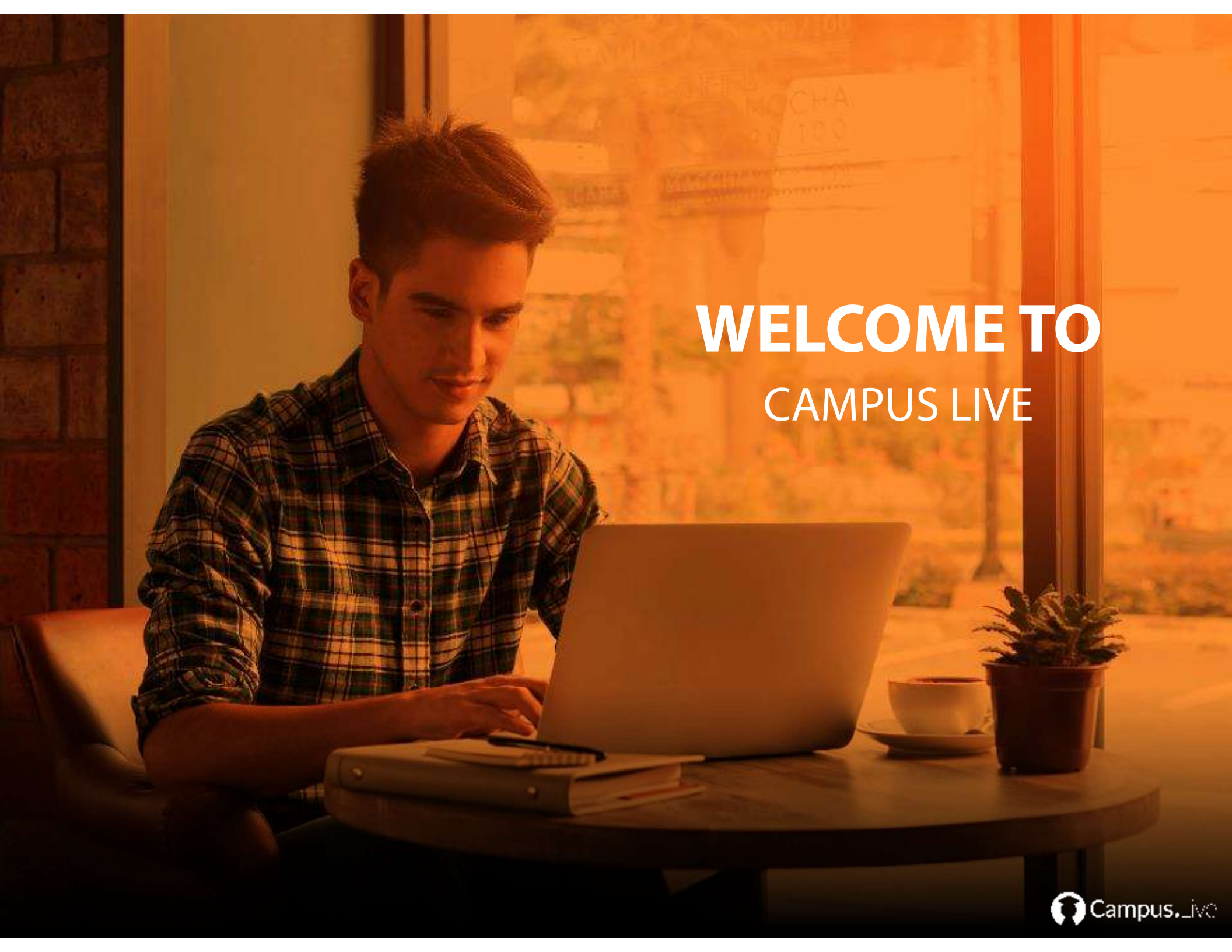
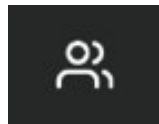




WELCOME TO CAMPUS LIVE

● USING THE PLATFORM

You will never appear on the screen.



Attendees

← To see the names of your classmates.

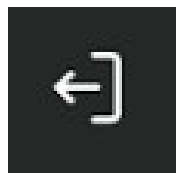


Chat

← To communicate with the class or teacher.

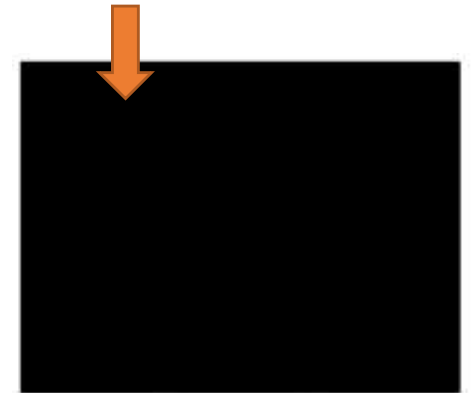


← To raise or lower your hand.



Exit

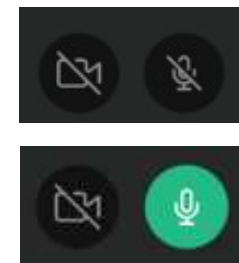
← To leave the class at the end.



Muted / unmuted.



Muted / unmuted.



→ Your camera will NEVER be turned on.

THIS LESSON IS ABOUT:

- RENEWABLE RESOURCES



Before Starting

- Tell us about you. Introduce yourself.
- Where do you live / work?
- What do you do for a living?
- What lesson have you enjoyed the most so far?
- Do you have any suggestions to share with us? Your opinion is really important for us.

Think of synonyms for the following words:

Energy

- Everlasting
- Free of cost
- Environmentally friendly
- Available
- Abundant
- Worldwide

Sources

- Toxic
- Hazard
- Fossil
- Costly
- Exhaustible
- Localized

What is Sustainable Energy?

Sustainable energy is a form of energy that meets today's _____ for it without putting it in danger of _____ and can be used over and over again. Sustainable energy does not cause any harm to the _____ and is available widely, _____. All renewable energy sources like solar, wind, _____, hydropower, nuclear, and _____ energy are sustainable as they are _____ (wind, sun, etc) and available in _____

*abundance – environment - stable - ocean – depletion - free of cost –
demand - geothermal*

Do you agree that sustainable energy is free of cost?

Why? / Why not?

Match the pictures with their corresponding names.

A



B



C



D



Geothermal
Energy

1

Ocean Energy

2

Wind Energy

3

Solar
Energy

4

Reading Part 1: *Solar and Geothermal Energy*

Solar energy is the energy that is produced by the sun in the form of heat and light. It is one of the most renewable and readily available sources of energy on planet Earth. The fact that it is available in plenty and free and does not belong to anybody makes it one of the most important of the non-conventional sources of energy. Solar energy has been used by people since ancient times by using simple magnifying glasses to concentrate the light of the sun into beams so hot they would cause wood to catch fire.

Geothermal energy is heat emanating from underneath the surface of the earth. It's also a green source of energy, meaning it does not emit greenhouse gasses that are hazardous to human and environmental health. When a volcano is active, lava erupts from underneath. This lava is partly magma. Underneath the earth's surface, the water and rocks absorb heat from the magma. As the depth increases, so do the temperatures of the underground water and rocks. Individuals across the world take advantage of the underground energy to heat their homes and generate electricity by digging up deep wells and subsequently pumping the hot underground water or steam to the earth's surface.

Comprehension check: Correct these statements:

- *Solar energy is Kinesthetic.*
- *It can be found in local areas.*
- *Geothermal energy is extracted from the surface of the Earth.*
- *Water and rocks receive heat from the atmosphere.*
- *The heat decreases as the depth increases.*

Reading Part 2: Ocean Energy

The oceans have proved to be a promising and renewable source of energy.

Wave energy: The wave energy is a form of kinetic energy. It is captured directly from the surface of the waves produced in the oceans. This is used to power a turbine. A turbine is used in a hydroelectric power station and functions based on the water conditions. **However**, the main issue with the ocean wave energy is that it is costly to set up. **Also**, water conditions are not the same in every part of the world.

Tidal energy: The movement of water currents uses the rise and fall of tides to generate electricity. It uses tidal energy generators beneath the sea which experience high tidal movements. The tidal energy generators capture the kinetic motion of the ebb and flow of ocean tides to produce electricity. **Since** tides are much more predictable than waves, it is **a better option** than wave energy. **But** to make it work, you need a large increase in tides. There are few places on earth that experience huge tides which makes its overall applicability a bit difficult. **Another issue** with this energy source is that building of huge dams and turbines may pose some effects on aquatic life.

Ocean thermal energy uses the difference in temperature on the surface and below to generate energy. The water gets too cold as you deep dive into the ocean. The greater the difference in temperature, the bigger will result in this method being more efficient.

Match the resources to their features.

Renewable resources

- Wave Energy
- Tidal Energy
- Ocean Thermal Energy

Characteristics

- It uses generators beneath the sea which experience high tidal movements.
- It is costly to set up.
- It uses the difference in temperature to produce energy.
- It comes from the movement on the surface of the waves.
- Conditions are not the same in every part of the world.



Reading Part 3: Wind and Nuclear Energy

Wind energy: the sun's rays unevenly heat the atmosphere. The entire heating process, in conjunction with the earth's rotation, produces wind that disperses throughout the different earth surfaces. The same wind can be harnessed to rotate turbines, consequently, generating electricity. It is the cleanest and most reliable way to generate electricity

Wind power describes the process by which wind is used to make the mechanical energy or electricity. Wind turbines convert the kinetic energy from wind energy into mechanical power. This mechanical power may be used for milling or pumping water, or a generator can convert mechanical energy to electrical energy for homes, businesses, schools, or other facilities.



Nuclear energy is the energy contained in the nucleus, or core of an atom. An **atom** is the building block of matter. The nucleus is held together by energy. Nuclear energy can be utilized to generate electricity. **However**, it must be released from the atom first. This calls for splitting of the atom. **Nuclear fission** is a process where atoms are split to discharge vast quantities of energy. **Even though** nuclear power is safer than burning electricity for fossil fuels, it is still a hazard to our health and the environment. **Despite** this energy being renewable, the uranium which is needed for atom splitting is not.

.

Expressing contrasts and expanding your argument

- Nuclear energy can be utilized to generate electricity. However, it must be released from the atom first.
- Tidal energy is a better option than wave energy. But to make it work, you need a large increase in tides.
- Even though nuclear power is safer than burning fossil fuels, it is still a hazard to our health and the environment.
- Also, water conditions are not the same in every part of the world.
- Another issue with this energy source is that building huge dams and turbines may pose some effects on aquatic life.
- Since tides are much more predictable than waves, it is a better option than wave energy.
- Despite this energy being renewable, the uranium, which is needed for atom splitting, is not.

Use the chart below to talk about renewable resources.

Energy	Availability	Low cost	Hazard	Environment friendly
Solar	In plenty	Free	None	Yes
Nuclear	Uranium is non-renewable	Nuclear reactors are costly	Radioactivity	No
Ocean	In abundance	Costly set up	None	May affect aquatic life.
Wind	In abundance	Free	None	Yes
Geothermal	In plenty	Processing plants are costly	None	Yes

Note: while the energy is free, capturing it is not - similarly to ocean and geothermal, capturing that energy IS costly.



Summing Up

What renewable resources are available in your country?

Today we learned...

- ✓ To talk about renewable resources.
- ✓ To scan a text (read to obtain specific information).
- ✓ Vocabulary connected with renewable resources.
- ✓ To express contrast and expand arguments.



● RENEWABLE RESOURCES



THANK YOU!!

info@campus.live